example, an eschar may suggest the diagnosis of *scrub typhus* or *rickettsialpox* **(Fig. A1-33A) (Chap. 192)** in the appropriate setting. In other illnesses (e.g., anthrax) **(Fig. A1-52) (Chap. S4)**, an ulcer or eschar may be the only skin manifestation.

FURTHER READING

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22 Fever of Unknown Origin

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DEFINITION

Clinicians commonly refer to any febrile illness without an initially obvious etiology as *fever of unknown origin* (FUO). Most febrile illnesses either resolve before a diagnosis can be made or develop distinguishing characteristics that lead to a diagnosis. The term *FUO* should be reserved for prolonged febrile illnesses without an established etiology despite intensive evaluation and diagnostic testing. This chapter focuses on FUO in the adult patient.

FUO was originally defined by Petersdorf and Beeson in 1961 as an illness of at least 3 weeks' duration with fever of ≥38.3°C (≥101°F) on two occasions and an uncertain diagnosis despite 1 week of inpatient evaluation. Since then, several modifications of the definition have been proposed to better reflect outpatient-based health care and to exclude immunocompromised patients, who require a different and more aggressive approach. To reduce heterogeneity between cohorts, inclusion of qualitative criteria of minimal diagnostic testing are necessary. Accordingly, FUO is now defined as:

1. Fever ≥38.3°C (≥101°F) on at least two occasions
2. Illness duration of at least 3 weeks
3. No known immunocompromised state
4. Diagnosis that remains uncertain after a thorough history-taking, physical examination, and the following obligatory investigations: determination of erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP) level; platelet count; leukocyte count and differential; measurement of levels of hemoglobin, electrolytes, creatinine, total protein, alkaline phosphatase, alanine aminotransferase, aspartate aminotransferase, lactate dehydrogenase, creatine kinase, ferritin, antinuclear antibodies, and rheumatoid factor; protein electrophoresis; urinalysis; blood cultures (n = 3); urine culture; chest x-ray; abdominal ultrasonography; and tuberculin skin test (TST) or interferon γ release assay (IGRA)

Closely related to FUO is *inflammation of unknown origin (IUO)*, which has the same definition as FUO, except for the body temperature criterion: IUO is defined as the presence of elevated inflammatory parameters (CRP or ESR) on multiple occasions for a period of at least 3 weeks in an immunocompetent patient with normal body temperature, for which a final explanation is lacking despite history-taking, physical examination, and the obligatory tests listed above. It has been shown in multiple cohorts that the causes for IUO and FUO are the same and that their workup should be identical. Therefore, for convenience, the term FUO will refer to both FUO and IUO within the remainder of this chapter.

ETIOLOGY AND EPIDEMIOLOGY

The prevalence of FUO is largely unknown and will likely vary between different geographic regions and different health care settings. In the 1990s, Japan reported a prevalence of 2.9% in admitted patients, and a recent Danish registry study reported 6220 patients in a 10-year time frame.

Table 22-1 summarizes the findings of large studies on FUO conducted over the past 20 years.

Since the 1960s, the range of FUO etiologies has evolved as a result of changes in the definition, local disease epidemiology, the widespread use of antibiotics, and especially the availability of new diagnostic techniques. The proportion of cases caused by intraabdominal abscesses and tumors, for example, has decreased because of earlier detection by CT and ultrasound. In addition, infective endocarditis is a less frequent cause because blood culture and echocardiographic techniques have improved. Conversely, some diagnoses, such as acute HIV infection or autoinflammatory diseases, were unknown six decades ago.

Data on the outcome of FUO are difficult to compare between studies because of variability in patient selection and variability in workup between cohorts. Study outcome strongly depends on study design and the origin of the cohort. The chance of a final diagnosis is higher when (older) time-based criteria are used versus when (newer) criteria including a standard basic workup are used for patient selection. The chance of remaining without a final diagnosis is two to five times higher in patients from Europe compared to patients from Asia.

TABLE 22-1 Etiology of FUO: Pooled Results of Large Studies Published in the Past 20 Years (2003–2023)

GEOGRAPHIC AREA	NO. OF COHORTS (INCLUSION PERIOD)	NO. OF PATIENTS	INFECTIONS, MEDIAN % (RANGE)	NONINFECTIOUS INFLAMMATORY DISEASES, MEDIAN % (RANGE)	MALIGNANCY, MEDIAN % (RANGE)	MISCELLANEOUS, MEDIAN % (RANGE)	NO DIAGNOSIS, MEDIAN % (RANGE)
Western Europe	12 (1995–2020)	2100	15.5 (4–36)	25 (17–33)	11 (3–30)	7.5 (0–16)	39.5 (26–54)
Other European and Turkey	15 (1984–2019)	1615	42 (26–74)	23 (12–38)	14 (4–19)	5 (2–18)	16 (0–35)
Middle East	3 (2009–2010)[a]	1289	66 (42–79)	15 (7–17)	7 (1–30)	1 (0–12)	8 (2–12)
Asia	39 (1994–2021)[a]	7191	42 (3–58)	19 (7–57)	13 (6–23)	6.5 (0–15)	17 (0–81)

Note: No studies from the Americas, Africa, or Oceania have been reported. Studies aimed at diagnostic methods were excluded.

[a]Some studies did not report an inclusion period.

For references, see supplementary material at *accessmedicine.com/harrisons*.

Roughly comparable to 60 years ago, infections remain the most common cause of FUO in non-Western cohorts. Compared to Europe, the risk of having an infection is over four times higher in patients from Southern Asia and three times higher in far East Asia. Up to half of all infections in patients with FUO outside Western nations are caused by *Mycobacterium tuberculosis*, which is a less common cause in Western Europe and probably also in the United States. Recent data on FUO from the latter, however, have not been reported. In Western cohorts, noninfectious inflammatory diseases (NIIDs), including autoimmune, autoinflammatory, and granulomatous diseases, as well as vasculitides, form the most common cause of FUO. Up to one-third of Western patients with FUO have a diagnosis that falls within the category of NIIDs. The number of FUO patients diagnosed with NIIDs probably will not decrease in the near future, as inflammation may precede more typical manifestations or laboratory evidence of these diseases by months. Moreover, NIIDs may only be diagnosed after prolonged observation and exclusion of other diseases, and an increasing number of genetic inflammatory syndromes have been described in more recent years.

Especially in Western cohorts, where the proportion of undiagnosed patients can be up to >50%, the proportion of patients remaining without a final diagnosis is much higher than 60 years ago despite technical advantages. This is called "the FUO paradox" and can be explained by the fact that in patients with fever, a diagnosis is often established before 3 weeks have elapsed because these patients tend to seek medical advice earlier and because diagnostic techniques such as CT, MRI, and positron emission tomography (PET)/CT, are now widely available. Therefore, only the cases that are most difficult to diagnose continue to meet the criteria for FUO. Furthermore, most patients who have FUO without a diagnosis currently do well. A less aggressive diagnostic approach may be used in clinically stable patients once diseases with immediate therapeutic or prognostic consequences have been ruled out.

No clinical parameters or tools that can help identify patients with a high risk of remaining undiagnosed have been identified. Recurrent inflammation (defined as repeated episodes of fever or inflammation interspersed with fever-free intervals of at least 2 weeks and apparent remission of the underlying disease) reduces the chance of finding a final explanation and is associated with a lower, but not zero, prevalence of infections. A symptom duration of >12 months is significantly associated with a lower chance of finding a final explanation, and infections and malignancy are less common, but still a possibility, in patients with such extensive symptom duration. It is important to note that even in patients with symptom duration >3 months, more common infections such as tuberculosis, spondylodiscitis, and endocarditis have been described and that lymphoma has been reported in patients with symptom duration >12 months. Normal PET/CT has also been associated with a lower chance of a final diagnosis. Artificial intelligence, using computer models to predict the final diagnosis in patients with FUO, may be able to shorten diagnostic delay and may guide specific diagnostic testing, but only limited data are available. Its exact value on a larger scale, especially in different geographic regions, is difficult to estimate at this moment.

DIFFERENTIAL DIAGNOSIS

The differential diagnosis for FUO is extensive. It is important to note that patients with FUO most often suffer from an atypical presentation of a more common disease, rather than from a rare disease. Unsurprisingly, the distribution of causes of FUO is influenced by local epidemiology. **Table 22-2** presents an overview of possible causes of FUO. Endocarditis, diverticulitis, vertebral osteomyelitis, and extrapulmonary tuberculosis are the more common infectious disease diagnoses. Q fever (*Coxiella burnetii*) and Whipple's disease (*Tropheryma whipplei* infection) are quite rare but should always be kept in mind as causes of FUO since the presenting symptoms can be nonspecific. Serologic testing for Q fever, which results from exposure to animals or animal products, should be performed by immunofluorescence assay (IFA) when the patient lives in a rural area or has a history of heart valve disease, an aortic aneurysm, or a vascular prosthesis. In patients with unexplained symptoms localized to the central nervous system, gastrointestinal tract, or joints, polymerase chain reaction testing for *T. whipplei* should be performed on stool and blood early in the diagnostic process. Travel to or (former) residence in tropical countries or the American Southwest should lead to consideration of endemic infectious diseases such as malaria, leishmaniasis, histoplasmosis, or coccidioidomycosis. Fever with signs of endocarditis and negative blood culture results poses a special problem. Culture-negative endocarditis **(Chap. 133)** may be due to difficult-to-culture bacteria such as nutritionally variant bacteria, HACEK organisms (including *Haemophilus parainfluenzae*, *H. paraphrophilus*, *Aggregatibacter actinomycetemcomitans*, *A. aphrophilus*, *A. paraphrophilus*, *Cardiobacterium hominis*, *C. valvarum*, *Eikenella corrodens*, and *Kingella kingae*), *Coxiella burnetii*, *T. whipplei*, and *Bartonella* species. Marantic endocarditis is a sterile thrombotic disease that occurs as a paraneoplastic phenomenon, especially with adenocarcinomas. Sterile endocarditis is also seen in the context of systemic lupus erythematosus and antiphospholipid syndrome. The use of next-generation sequencing for rapid detection of infecting microbes may be valuable in patients with FUO, but its exact value and exact application are yet to be determined.

Of the NIIDs, adult-onset Still's disease, large-vessel vasculitis, polymyalgia rheumatica, systemic lupus erythematosus (SLE), and sarcoidosis are rather common diagnoses in patients with FUO. Autoinflammatory syndromes are rare (with the exception of familial Mediterranean fever in specific geographic regions), and most can only be diagnosed on clinical criteria. Schnitzler's syndrome, which can present at any age, can often be diagnosed easily in a patient with FUO who presents with urticaria, bone pain, and monoclonal gammopathy. A new autoinflammatory syndrome called vacuoles, E1 enzyme, X-linked autoinflammatory syndrome (VEXAS) has been described in >300 patients since 2020. VEXAS is an adult-onset multisystem autoinflammatory syndrome with predominance in middle-aged to elderly males and is characterized, among others, by recurrent inflammation, skin lesions, chondritis, lung disease, venous thrombosis, arthritis, and myelodysplastic syndrome. The presence of vacuoles in myeloid precursors in bone marrow biopsy is very suspicious for VEXAS in a patient with inflammatory symptoms. The disease is caused by somatic mosaicism for pathogenic variants in *UBA1*.

Although most malignancies can present with fever, malignant lymphoma is by far the most common diagnosis of FUO among the neoplasms. Sometimes the fever even precedes lymphadenopathy detectable by physical examination.

Apart from drug-induced fever and exercise-induced or benign hyperthermia, none of the miscellaneous causes of fever is found very frequently in patients with FUO. Virtually all drugs can cause fever, even after long-term use. *Drug-induced fever*, including DRESS (drug reaction with eosinophilia and systemic symptoms; now more generally referred to as drug-induced hypersensitivity syndrome [DIHS]; **Fig. A1-48**), is often accompanied by eosinophilia and also by lymphadenopathy, which can be extensive. More common causes of drug-induced fever are allopurinol, carbamazepine, lamotrigine, phenytoin, sulfasalazine, furosemide, and antimicrobial drugs (especially sulfonamides, minocycline, vancomycin, β-lactam antibiotics, and isoniazid). Benign hyperthermia or *exercise-induced hyperthermia* **(Chaps. 20 and 478)** is characterized by an elevated body temperature without an increase in CRP or ESR or other signs of inflammation; these may be postinfectious in origin. Infectious triggers for benign hyperthermia are possibly the same as for chronic fatigue syndrome, such as Epstein-Barr virus, Q-fever, and COVID-19. *Factitious fever* (fever artificially induced by the patient—for example, by IV injection of contaminated water) should be considered in all patients, especially those with easy access to medical equipment. In *fraudulent fever*, the patient is normothermic but manipulates the thermometer. Simultaneous measurements at different body sites (rectum, ear, mouth) should rapidly identify this diagnosis. Another clue to fraudulent fever is dissociation between pulse rate and temperature.

TABLE 22-2 Reported Causes of Fever of Unknown Origin (FUO)[a]

Infections	
Bacterial, nonspecific	Abdominal abscess, adnexitis, aortitis, apical granuloma, appendicitis, bacterial translocation, bronchiectasis, cholangitis, cholecystitis, diverticulitis, endocarditis, endometritis, epididymitis, epidural abscess, infected joint prosthesis, infected vascular catheter, infected vascular prosthesis, infectious arthritis, infective myonecrosis, intracranial abscess, liver abscess, lung abscess, malakoplakia, mastitis, mastoiditis, mediastinitis, muscle abscess, mycotic aneurysm, osteomyelitis, pelvic inflammatory disease, prostatitis, pyelonephritis, pylephlebitis, pyomyoma, renal abscess, septic arthritis, septic phlebitis, sinusitis, spondylodiscitis, xanthogranulomatous urinary tract infection
Bacterial, specific	Actinomycosis, *Bacteroides*, bartonellosis, brucellosis, Bacillus Calmette-Guérin infection, *Burkholderia* spp., *Campylobacter* infection, *Chlamydia pneumoniae* infection, chronic meningococcemia, ehrlichiosis, enterococci, *Escherichia coli*, *Gemella* infection, gonococcus, *Klebsiella* spp., *Lactobacillus*, legionellosis, leptospirosis, listeriosis, louse-borne relapsing fever (*Borrelia recurrentis*), Lyme disease (*Borrelia burgdorferi*), melioidosis (*Burkholderia pseudomallei*), *Mycoplasma* infection, nocardiosis, nontuberculous mycobacteria, *Prevotella* infection, *Pseudomonas* spp., psittacosis, Q fever (*Coxiella burnetii*), rickettsiosis, *Rhodococcus* spp., *Salmonella* spp., *Spirillum minor* infection, *Sphingobacterium* infection, *Staphylococcus aureus, Streptobacillus moniliformis* infection, syphilis, tick-borne relapsing fever (*Borrelia duttonii, Borrelia hermsi*), tuberculosis, tularemia, typhoid fever and other salmonelloses, Whipple disease (*Tropheryma whipplei*), yersiniosis
Fungal	Aspergillosis, blastomycosis, candidiasis, coccidioidomycosis, cryptococcosis, histoplasmosis, *Malassezia furfur* infection, mucormycosis, paracoccidioidomycosis, *Penicillium* spp., *Pneumocystis jirovecii* pneumonia, sporotrichosis, *Talaromyces* infection
Parasitic	Amebiasis, babesiosis, echinococcosis, fascioliasis, malaria, schistosomiasis, strongyloidiasis, toxocariasis, toxoplasmosis, trichinellosis, trypanosomiasis, visceral leishmaniasis
Viral	Colorado tick fever, coxsackievirus infection, cytomegalovirus infection, chikungunya, dengue, Epstein-Barr virus infection, hantavirus infection, hepatitis (A, B, C, D, E), herpes simplex, HIV infection, human herpesvirus 6 infection, parvovirus infection, West Nile virus infection
Noninfectious Inflammatory Diseases	
Systemic rheumatic and autoimmune diseases	Ankylosing spondylitis, antiphospholipid syndrome, autoimmune hemolytic anemia, autoimmune hepatitis, Behçet's disease, cryoglobulinemia, dermatomyositis, Felty syndrome, gout, mixed connective tissue disease, neuromyelitis optica, polymyositis, pseudogout, reactive arthritis, psoriatic arthritis, relapsing polychondritis, rheumatic fever, rheumatoid arthritis, Sjögren's syndrome, systemic lupus erythematosus, systemic sclerosis, Vogt-Koyanagi-Harada syndrome
Vasculitis	Allergic vasculitis, antineutrophil cytoplastic antibody (ANCA) vasculitis, Cogan's syndrome, eosinophilic granulomatosis with polyangiitis, giant cell vasculitis/polymyalgia rheumatica, granulomatosis with polyangiitis, hypersensitivity vasculitis, IgA vasculitis, Kawasaki disease, polyarteritis nodosa, Takayasu arteritis, urticarial vasculitis
Granulomatous diseases	Idiopathic granulomatous hepatitis, sarcoidosis
Autoinflammatory syndromes	Blau syndrome, CAPS[b] (cryopyrin-associated periodic syndromes), Crohn's disease, DIRA (deficiency of the interleukin 1 receptor antagonist), deficiency of adenosine deaminase 2 (DADA2), familial Mediterranean fever, hyper-IgD syndrome (HIDS, due to mevalonate kinase deficiency), juvenile idiopathic arthritis, macrophage activation syndrome, NLRP12-related disease, PAPA syndrome (pyogenic sterile arthritis, pyoderma gangrenosum, and acne), PFAPA syndrome (periodic fever, aphthous stomatitis, pharyngitis, adenitis), recurrent idiopathic pericarditis, SAPHO (synovitis, acne, pustulosis, hyperostosis, osteomyelitis), Schnitzler's syndrome, Still's disease, TRAPS (tumor necrosis factor receptor–associated periodic syndrome), VEXAS (vacuoles, E1 enzyme, X-linked autoinflammatory syndrome)
Neoplasms	
Hematologic malignancies	Amyloidosis, angioimmunoblastic lymphoma, Castleman's disease, hemophagocytic syndrome (hemophagocytic lymphohistiocytosis), Hodgkin's disease, hypereosinophilic syndrome, leukemia, lymphomatoid granulomatosis, malignant histiocytosis, multiple myeloma, myelodysplastic syndrome, myelofibrosis, non-Hodgkin's lymphoma, plasmacytoma, systemic mastocytosis, vaso-occlusive crisis in sickle cell disease
Solid tumors	Most solid tumors and metastases can cause fever. Those most common causing FUO are breast, colon, hepatocellular, lung, pancreatic, and renal cell carcinomas.
Benign tumors	Angiomyolipoma, cavernous hemangioma of the liver, craniopharyngioma, necrosis of dermoid tumor in Gardner's syndrome, tumoral calcinosis
Miscellaneous Causes	
	ADEM (acute disseminated encephalomyelitis), adrenal insufficiency, alcoholic steatohepatitis, allergic purpura, aneurysms, anal fistula, anti-NMDA encephalitis, antisynthetase syndrome, anomalous thoracic duct, aortic dissection, aortic-enteral fistula, aseptic meningitis (Mollaret's syndrome), atrial myxoma, brewer's yeast ingestion, calcium pyrophosphate deposition, Caroli's disease, cholesterol emboli, cirrhosis, complex partial status epilepticus, crowned dense syndrome, cyclic neutropenia, cryptogenic organizing pneumonia, drug fever, Erdheim-Chester disease, extrinsic allergic alveolitis, Fabry's disease, factitious disease, fire-eater's lung, fraudulent fever, Dressler's syndrome, ganglioneuroma, extrinsic allergic alveolitis, Gaucher's disease, hemolytic uremic syndrome, Hamman-Rich syndrome (acute interstitial pneumonia), Hashimoto's encephalopathy, hematoma, heparin-induced thrombocytopenia, histiocytic necrotizing lymphadenitis, hypersensitivity pneumonitis, hypertriglyceridemia, hypogammaglobulinemia/subclass deficiency, hypothalamic hypopituitarism, idiopathic inflammatory myopathy, idiopathic normal-pressure hydrocephalus, IgG4 disease, inflammatory pseudotumor, interstitial nephritis, Kikuchi's disease, limbic encephalitis, linear IgA dermatosis, mesenteric fibromatosis, metal fume fever, microaspiration, milk protein allergy, Mollaret's meningitis, myotonic dystrophy, nonbacterial osteitis, obstructive suppurative pancreatic ductitis, organic dust toxic syndrome, panniculitis, POEMS (polyneuropathy, organomegaly, endocrinopathy, monoclonal protein, skin changes), obstructive sleep apnea syndrome, organizing pneumonia, polymer fume fever, post–cardiac injury syndrome, primary biliary cirrhosis, primary hyperparathyroidism, pseudomembranous colitis, pulmonary embolism, pulmonary nodular lymphoid hyperplasia, pyoderma gangrenosum, recurrent large bowel ischemia, retroperitoneal fibrosis, Rosai-Dorfman disease, sclerosing mesenteritis, silicone embolization, subacute thyroiditis (de Quervain's), Sweet's syndrome (acute febrile neutrophilic dermatosis), thrombosis, TAFRO syndrome (thrombocytopenia, anasarca, fever, reticulin fibrosis, and organomegaly) tubulointerstitial nephritis and uveitis syndrome (TINU), ulcerative colitis, XIAP deficiency (X-linked inhibitor of apoptosis deficiency)
Thermoregulatory Disorders	
Central	Brain tumor, cerebrovascular accident, encephalitis, hypothalamic dysfunction
Peripheral	Anhidrotic ectodermal dysplasia, benign hyperthermia, exercise-induced hyperthermia, heat stroke, hyperthyroidism, paroxysmal sympathetic hyperactivity, pheochromocytoma

[a]This table includes causes of FUO that have been described in the literature. [b]CAPS includes chronic infantile neurologic cutaneous and articular syndrome (CINCA, also known as neonatal-onset multisystem inflammatory disease, or NOMID), familial cold autoinflammatory syndrome (FCAS), and Muckle-Wells syndrome.

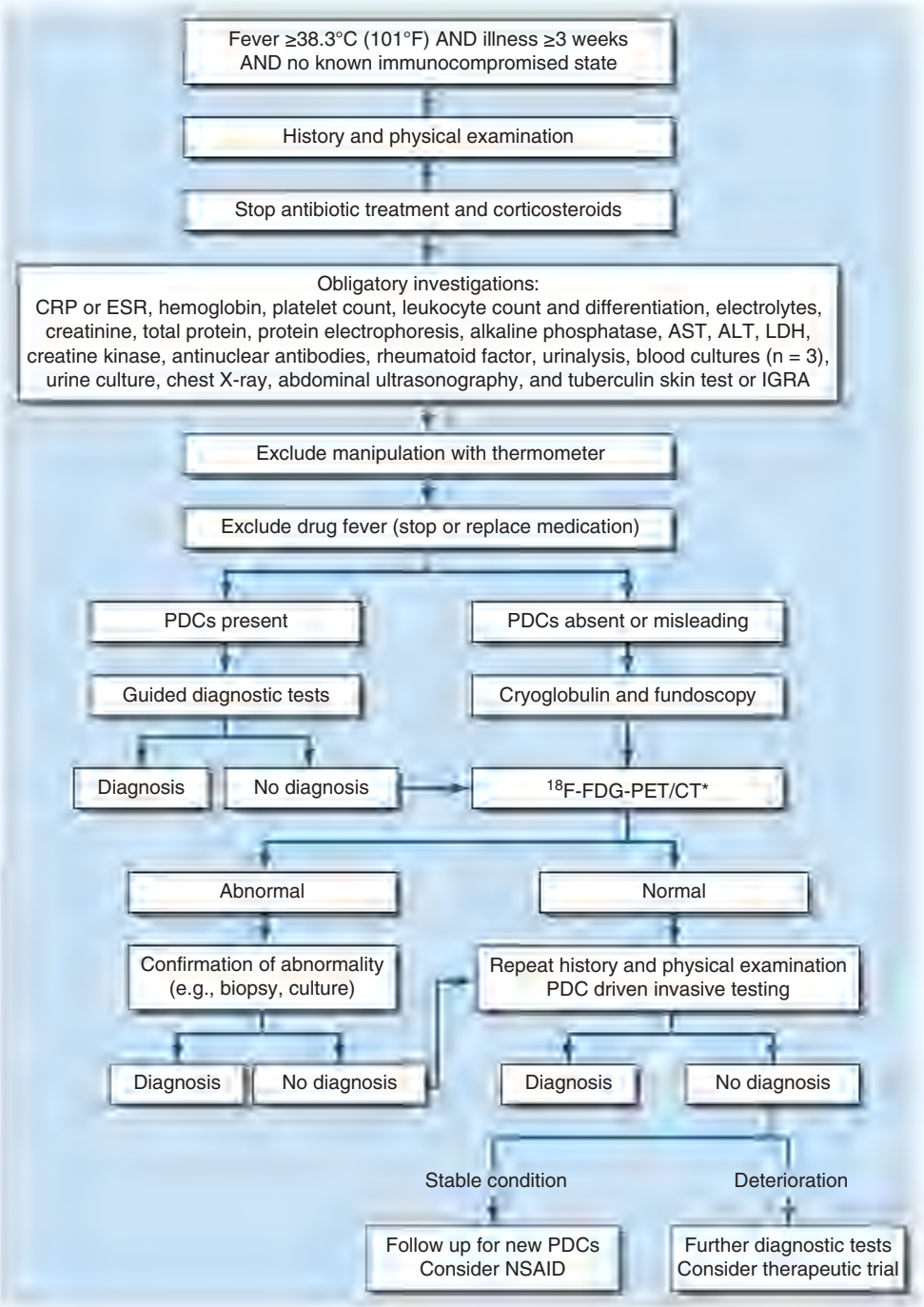

FIGURE 22-1 Structured approach to patients with FUO. ALT, alanine aminotransferase; AST, aspartate aminotransferase; CRP, C-reactive protein; ESR, erythrocyte sedimentation rate; ^{18}F-FDG-PET/CT, ^{18}F-fluorodeoxyglucose positron emission tomography combined with low-dose CT; IGRA, interferon γ release assay; LDH, lactate dehydrogenase; NSAID, nonsteroidal anti-inflammatory drug; PDCs, potentially diagnostic clues (all localizing signs, symptoms, and abnormalities potentially pointing toward a diagnosis). *If not available, gallium scintigraphy or labeled leukocyte scintigraphy is an acceptable alternative.

APPROACH TO THE PATIENT

A Structured Diagnostic Approach

Figure 22-1 shows a structured approach to patients presenting with FUO. The most important step in the diagnostic workup is the search for potentially diagnostic clues (PDCs) through complete and repeated history-taking and physical examination and the obligatory investigations listed above and in the figure. PDCs are defined as all localizing signs, symptoms, and abnormalities potentially pointing toward a diagnosis. Although PDCs are often misleading, only with their help can a concise list of probable diagnoses be made. The history should include information about the fever pattern (continuous or recurrent) and duration, previous medical history, present and recent drug use, family history, sexual history, country of origin, recent and remote travel, environmental exposures associated with travel or hobbies, and animal contacts. A complete physical examination should be performed, with special attention to the eyes, lymph nodes, temporal arteries, liver, spleen, sites of previous surgery, entire skin surface, and mucous membranes. Before further diagnostic tests are initiated, antibiotic and glucocorticoid treatment, which can mask many diseases, should be stopped. For example, sterile blood and other cultures are not completely reliable when samples are obtained during antibiotic treatment, and the size of enlarged lymph nodes usually decreases during glucocorticoid treatment, regardless of the cause of lymphadenopathy. Despite the high percentage of false-positive

ultrasounds and the relatively low sensitivity of chest x-rays, the performance of these simple, low-cost diagnostic tests remains obligatory in all patients with FUO in order to separate cases that are caused by easily diagnosed diseases from those that are not. Abdominal ultrasound and conventional chest radiography have been preferred to thoracic and abdominal CT as an obligatory test because of their relatively low cost, lower radiation burden, and absence of side effects.

After identification of all PDCs retrieved from the history, physical examination, and obligatory tests, a limited list of the most probable diagnoses should be made. Since most investigations are helpful only for patients who have PDCs for the diagnoses sought, further diagnostic procedures should be limited to specific investigations aimed at confirming or excluding diseases on this list. In FUO, the diagnostic pointers are numerous and diverse but may be missed on initial examination, often being detected only by subsequent careful examinations. In the absence of PDCs, the history and physical examination should be repeated regularly. In patients with recurrent inflammation, the diagnostic workup is the same as for patients with continuous inflammation. Patients should be asked to return during a febrile episode so that the history, physical examination, and laboratory tests can be repeated during a symptomatic phase. Further diagnostic tests should be performed only during an inflammatory episode because abnormalities may be absent between episodes.

One of the first steps should be to rule out benign hyperthermia or factitious or fraudulent fever, particularly in patients without signs of inflammation in laboratory tests. All medications, including nonprescription drugs and nutritional supplements, should be discontinued early in the evaluation to exclude drug fever. If fever persists beyond 72 h after discontinuation of the suspected drug, it is unlikely that this drug is the cause.

Only rarely do biochemical tests (beyond the obligatory tests needed to classify a patient's fever as FUO) lead directly to a definitive diagnosis in the absence of PDCs. The diagnostic yield of immunologic serologies other than those included in the obligatory tests is relatively low. These tests more often yield false-positive rather than true-positive results and are of little use without PDCs pointing to specific disorders. Given the absence of specific symptoms in many patients and the relatively low cost of the test, investigation of cryoglobulins appears to be a valuable screening test in patients with FUO.

Multiple blood samples should be cultured in the laboratory long enough to ensure ample time for any fastidious organisms, such as those of the HACEK group. It is critical to inform the laboratory of the intent to test for unusual organisms. Specialized media should be used when the history suggests uncommon microorganisms, such as *Histoplasma* or *Legionella*. Performing more than three blood cultures or more than one urine culture is useless in patients with FUO in the absence of PDCs (e.g., a high level of clinical suspicion of endocarditis). Repeating blood or urine cultures is useful only when previously cultured samples were collected during antibiotic treatment or within 1 week after its discontinuation. FUO with headache should prompt microbiologic examination of cerebrospinal fluid (CSF) for organisms including herpes simplex virus (especially type 2), *Cryptococcus neoformans* and *C. gattii*, and *Mycobacterium tuberculosis*.

Microbiologic serology should not be included in the diagnostic workup of patients without PDCs for specific infections. A tuberculin skin test (TST) or interferon γ (IFNγ) release assay (IGRA) is included in the obligatory investigations, but it may yield indeterminate results in patients with miliary tuberculosis, with malnutrition, receiving immunosuppression, or with anti-IFNγ autoantibodies. Although the IGRA is less influenced by prior vaccination with bacille Calmette-Guérin (BCG) or by infection with most nontuberculous mycobacteria, its sensitivity is similar to that of the TST; a negative TST or IGRA does not exclude a diagnosis of tuberculosis. Miliary tuberculosis is especially difficult to diagnose. Granulomatous disease in liver or bone marrow biopsy samples, for example, should always lead to a (re)consideration of this diagnosis. If miliary tuberculosis is suspected, liver biopsy for acid-fast smear, culture, and polymerase chain reaction probably still has the highest diagnostic yield; however, biopsies of bone marrow, lymph nodes, or other involved organs also can be considered. Mycobacterial polymerase chain reaction (PCR) testing together with cultures may shorten the time to diagnosis compared to mycobacterial cultures alone.

The diagnostic yield of echocardiography, sinus radiography, radiologic or endoscopic evaluation of the gastrointestinal tract, bone marrow biopsy, liver biopsy, and bronchoscopy is very low in the absence of PDCs.

In patients without PDCs or with only misleading PDCs, fundoscopy by an ophthalmologist may be useful early in the diagnostic workup to exclude retinal vasculitis.

Several studies have shown a high prevalence of giant cell arteritis among patients with FUO, with rates up to 17% among elderly patients. Giant cell arteritis often involves large arteries and, in most cases, can be diagnosed by ^{18}F-fluordeoxyglucose (^{18}F-FDG) positron emission tomography (PET) with combined CT. However, temporal artery biopsy is still recommended for patients ≥55 years. ^{18}F-FDG-PET/CT may be confounded in vasculitis limited to the temporal arteries because of the small diameter of these vessels and the high levels of FDG uptake in the brain.

When PDC-guided diagnostic tests do not lead to a diagnosis, ^{18}F-FDG-PET/CT should be performed, preferably at a time point with elevated ESR or CRP to increase the chance of positive findings. This is especially important in patients with episodic FUO.

^{18}F-FLUORDEOXYGLUCOSE POSITRON EMISSION TOMOGRAPHY

^{18}F-FDG-PET has become an established imaging procedure in FUO. FDG accumulates in tissues with a high rate of glycolysis, which occurs not only in malignant cells but also in activated leukocytes and thus permits the imaging of acute and chronic inflammatory processes. Compared with conventional scintigraphy (see below), ^{18}F-FDG-PET/CT offers the advantages of higher resolution, greater sensitivity in chronic low-grade infections, and a high degree of accuracy in the central skeleton. Furthermore, vascular uptake of FDG is increased in patients with vasculitis **(Fig. 22-2)**. The mechanisms responsible for FDG uptake do not allow differentiation among infection, sterile inflammation, and malignancy, but because all of these can cause FUO, this is an overall advantage. It is important to realize that physiologic uptake or concentration of FDG may obscure pathologic foci in the brain, heart, bowel, kidneys, and bladder. FDG uptake in the heart, which obscures endocarditis, may be prevented by consumption of a low-carbohydrate diet before the PET investigation. FDG uptake in the brain may obscure local temporal large vessel vasculitis. In patients with fever, bone marrow uptake is frequently increased in a nonspecific way due to cytokine activation, which upregulates glucose transporters in bone marrow cells.

In recent years, many cohort studies and several meta-analyses have focused on the diagnostic yield of PET/CT in FUO. These studies are highly variable in terms of the selection of patients, the follow-up, and the selection of a gold-standard reference. Indirect comparisons of test performance suggested that ^{18}F-FDG-PET/CT outperformed gallium scintigraphy and leukocyte scintigraphy. Similarly, indirect comparisons of diagnostic yields suggested that ^{18}F-FDG-PET/CT was more likely than alternative tests to correctly identify the cause of FUO. Meta-analyses report a high diagnostic yield of PET/CT in the workup of FUO patients, with total diagnostic yield (i.e., the proportion of scans helpful in finding the final explanation for the inflammation) of ~50% for PET/CT.

As many patients with FUO present with periodic fever, the correct timing of PET/CT increases its diagnostic value. Few studies on the use of biomarkers, such as elevated CRP or ESR, as contributors to outcome of PET/CT have been performed. When both CRP and ESR are normal at the time of ^{18}F-FDG-PET/CT, the outcome may only be contributory if a patient does have fever at the time of the scan.

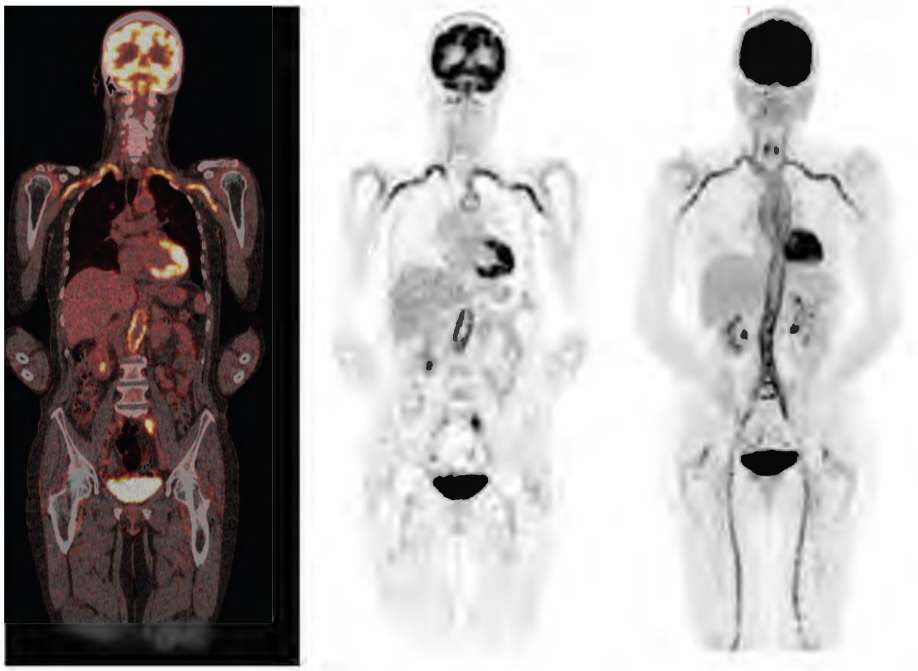

FIGURE 22-2 ^{18}F-FDG-PET/CT in a patient with FUO. This 72-year-old woman presented with a low-grade fever and severe fatigue of almost 3 months' duration. An extensive history was taken, but the patient had no specific complaints and had not traveled recently. Her previous history was unremarkable, and she did not use any drugs. Physical examination, including palpation of the temporal arteries, yielded completely normal results. Laboratory examination showed normocytic anemia, a C-reactive protein level of 43 mg/L, an erythrocyte sedimentation rate of 87 mm/h, and mild hypoalbuminemia. Results of the other obligatory tests were all normal. ^{18}F-FDG-PET/CT showed increased FDG uptake in all major arteries (carotid, jugular, and subclavian arteries; thoracic and abdominal aorta; iliac, femoral, and popliteal arteries) and in the soft tissue around the shoulders, hips, and knees—findings compatible with large-vessel vasculitis and polymyalgia rheumatica. Within 1 week after the initiation of treatment with prednisone (60 mg once daily), the patient completely recovered. After 1 month, the prednisone dose was slowly tapered.

Although PET/CT and other scintigraphic techniques do not directly provide a definitive diagnosis (with the exception of some patients with, for instance, large vessel vasculitis), they often identify the anatomic location of a particular ongoing metabolic process. With the help of other techniques such as biopsy and culture, a timely diagnosis and treatment can be facilitated. Pathologic FDG uptake is quickly eradicated by treatment with glucocorticoids in many diseases, including vasculitis and lymphoma; therefore, glucocorticoid use should be stopped or postponed until after ^{18}F-FDG-PET/CT is performed.

^{18}F-FDG-PET/CT is a relatively expensive procedure whose availability is still limited in some regions. Nevertheless, ^{18}F-FDG-PET/CT can be cost-effective in the FUO diagnostic workup if used early, helping to establish an early diagnosis, reducing days of hospitalization for diagnostic purposes, and obviating unnecessary and unhelpful tests. When ^{18}F-FDG-PET/CT has been made under the right conditions (i.e., during active inflammation) but has not contributed to the final diagnosis, repeating PET/CT is probably of little value, unless new signs or symptoms appear.

ALTERNATIVES TO POSITRON EMISSION TOMOGRAPHY/ COMPUTED TOMOGRAPHY

When PET/CT is unavailable, other whole-body imaging modalities should be considered. Compared to CT, scintigraphy provides the location of active inflammation. Conventional scintigraphic methods in clinical practice are ^{67}Ga-citrate scintigraphy and ^{111}In- or ^{99m}Tc-labeled leukocyte scintigraphy. The diagnostic yields of these conventional scintigraphic studies are lower than for PET/CT: the diagnostic yield of gallium scintigraphy ranges from 21% to 54%, and on average, the location of a source of fever can correctly be localized in approximately one-third of patients. The diagnostic value of leukocyte scintigraphy ranges from 8% to 31%, and overall, the cause of FUO can correctly be identified in one-fifth of patients. Recently, one study of full-body MRI as an alternative for PET/CT reported a diagnostic yield comparable to PET/CT. Further studies are needed to confirm this finding.

LATER-STAGE DIAGNOSTIC TESTS

Later-stage diagnostic testing should always be PDC driven and may come from repeated history-taking, physical examination, or PET/CT. Biopsies for pathology and/or microbiology evaluation will often be necessary.

Diagnostic delay frequently results from a failure to recognize PDCs in the available information. When all PDCs have been thoroughly evaluated but no final explanation has been found, waiting for new PDCs to appear probably is better than ordering more screening investigations. Only when a patient's condition deteriorates without providing new PDCs should further diagnostic workup be performed.

SECOND OPINION IN AN EXPERT CENTER

When no explanation for FUO is found despite the workup described above, a second opinion from an FUO expert center should be considered. The single study on the value of second opinion in FUO reported that in over half of patients with unexplained FUO, a diagnosis could be found in an expert center. Additionally, of all patients who remained without a diagnosis after second opinion, 1 in 10 became fever-free upon empirical treatment, adding up to a beneficial outcome in a little under 70% of patients. One of the reasons for the higher diagnostic rate in expert centers is the fact that FUO is often an atypical presentation of a common disease, making pattern recognition informed by high exposure an important diagnostic tool.

TREATMENT

Fever of Unknown Origin

Rational treatment is based on the final diagnosis. In FUO patients remaining with unexplained FUO, the large majority may spontaneously become symptom-free. Empirical therapeutic trials should be avoided, except in cases in which a patient's condition is rapidly deteriorating. Drugs that are commonly used for therapeutic trials are antibiotics, antituberculars, nonsteroidal anti-inflammatory drugs, colchicine, and interleukin-1 inhibitors.

ANTIBIOTICS AND ANTITUBERCULOUS THERAPY

Antibiotic or antituberculous therapy may irrevocably diminish the ability to culture bacteria. However, hemodynamic instability or neutropenia is a good indication for empirical antibiotic therapy. If the TST or IGRA is positive, or if granulomatous disease is present with anergy and sarcoidosis seems unlikely, or in any patient coming from an endemic region with a clinical picture fitting extrapulmonary tuberculosis, a trial of antituberculous therapy may be started, but not before mycobacterial cultures and, if available, mycobacterial PCR testing have been performed on material collected from the suspected location of inflammation. Especially in miliary tuberculosis, it may be very difficult to obtain a rapid diagnosis. If the fever does not respond after 6 weeks of empirical antituberculous treatment, another diagnosis should be considered.

COLCHICINE, NONSTEROIDAL ANTI-INFLAMMATORY DRUGS, AND GLUCOCORTICOIDS

If the fever persists and the source remains elusive after completion of investigations, supportive treatment with acetaminophen or nonsteroidal anti-inflammatory drugs (NSAIDs) can be helpful. The response of Still's disease to NSAIDs is dramatic in some cases. Colchicine is highly effective in preventing attacks of familial Mediterranean fever (FMF) but is not always effective once an attack is underway. When FMF is suspected, the response to colchicine is not a completely reliable diagnostic tool in the acute phase, but within weeks to months of continuous colchicine treatment, most patients show remarkable improvements in the frequency and severity of subsequent febrile episodes. Therefore, colchicine may be tried in patients with features compatible with FMF, especially when these patients originate from a high-prevalence region. In patients suffering from pericarditis as one of the main associated symptoms, colchicine may also be effective to prevent recurrent attacks. If Behçet's disease is considered likely, colchicine may also have favorable effect.

The effects of glucocorticoids on giant cell arteritis and polymyalgia rheumatica are impressive. Early empirical trials with glucocorticoids, however, decrease the chances of reaching a diagnosis for which more specific, less morbid, and sometimes life-saving treatment might be more appropriate, such as malignant lymphoma. The ability of glucocorticoids to mask fever while permitting the spread of infection or lymphoma dictates that their use should be avoided unless infectious diseases and malignant lymphoma have been sufficiently ruled out and inflammatory disease is probable and is likely to be debilitating or threatening.

INTERLEUKIN-1 INHIBITION

Interleukin (IL) 1 is a key cytokine in local and systemic inflammation and the febrile response. The availability of specific IL-1-targeting agents has revealed a pathologic role of IL-1-mediated inflammation in a growing list of diseases. Anakinra, a recombinant form of the naturally occurring IL-1 receptor antagonist (IL-1Ra), blocks the activity of both IL-1α and IL-1β. Anakinra is extremely effective in the treatment of many autoinflammatory syndromes, such as FMF, cryopyrin-associated periodic syndrome, tumor necrosis factor receptor–associated periodic syndrome, mevalonate kinase deficiency (hyper-IgD syndrome), Schnitzler's syndrome, and Still's disease. There are many other chronic inflammatory disorders in which anti-IL-1 therapy is highly effective. A therapeutic trial with anakinra can be considered in patients whose FUO has not been diagnosed after later-stage diagnostic tests and show signs of IL-1-driven inflammation, such as serositis, elevated CRP, and elevated ferritin. When autoinflammation is considered in the differential diagnosis, IL-1 inhibition is preferred over corticosteroids to prevent the metabolic, immunologic, and gastrointestinal side effects of glucocorticoid administration, and because IL-1 inhibition has superior efficacy.

PROGNOSIS

The prognosis of patients with FUO mostly depends on the underlying disease. In patients in whom FUO remains unexplained despite extensive evaluation, the prognosis is favorable. The risk of FUO-related mortality is probably highest during the early phases of the diagnostic process: in a cohort study including 168 patients without a final diagnosis, all four patients who died did so during the index admission; in two of them, diagnoses were made upon autopsy (intravascular lymphoma and bilateral pneumonia).

Large cohort studies in patients remaining without a diagnosis report high percentages of spontaneous resolution of fever and a mortality of 8% or less during several years of follow-up. ^{18}F-FDG-PET/CT may be helpful to predict which patients will resolve because normal ^{18}F-FDG-PET/CT scans are associated with higher rates of spontaneous resolution.

FURTHER READING

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Section 3 Nervous System Dysfunction

23 Syncope

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Syncope is a transient, self-limited loss of consciousness due to acute global impairment of cerebral blood flow. The onset is rapid, duration brief, and recovery spontaneous and complete. Other causes of transient loss of consciousness need to be distinguished from syncope; these include seizures, vertebrobasilar ischemia, hypoxemia, and hypoglycemia. A syncopal prodrome (*presyncope*) is common, although loss of consciousness may occur without any warning symptoms. Typical presyncopal symptoms include lightheadedness or faintness, dizziness, weakness, fatigue, and visual and auditory disturbances. The causes of syncope can be divided into three general categories: (1) neurally mediated syncope (also called *reflex or vasovagal syncope*), (2) orthostatic hypotension, and (3) cardiac syncope.

Neurally mediated syncope comprises a heterogeneous group of disorders that are characterized by a transient change in the reflexes responsible for maintaining cardiovascular homeostasis. Episodic vasodilation (or loss of vasoconstrictor tone), decreased cardiac output, and bradycardia occur in varying combinations, resulting in temporary failure of blood pressure control. In contrast, in patients with orthostatic hypotension due to autonomic failure, these cardiovascular homeostatic reflexes are chronically impaired. Cardiac syncope may be due to arrhythmias or structural cardiac diseases that cause a decrease in cardiac output. The clinical features, underlying pathophysiologic mechanisms, therapeutic interventions, and prognoses differ markedly among these three causes.

EPIDEMIOLOGY AND NATURAL HISTORY

Syncope is a common presenting problem, accounting for ~3% of all emergency department (ED) visits and 1% of all hospital admissions. The annual cost for syncope-related hospitalization in the United States